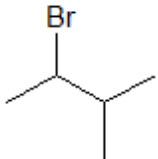


Section A

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
1						1		1		
2				$\text{Mg(s)} + \text{C(s)} + 1\frac{1}{2}\text{O}_2\text{(g)} \rightarrow \text{MgCO}_3\text{(s)}$ award (1) for reactants and product award (1) for balancing and state symbols - only if reactants and products correct	2			2		
3				award (1) for either of following • B can exist in <i>E-Z</i> forms because each of the double bonded carbon atoms has two different groups attached to it • A cannot exist in <i>E-Z</i> forms because (one of) the double bonded carbon atoms has two groups attached to it which are the same B is Z-but-2-ene (1) award (1) for 2-methylpropene if isomer A chosen		2		2		
4				$\text{C}_3\text{H}_7\text{Br}$ is hydrolysed most rapidly (1) because the C—Br bond is the weakest (of the C-halogen) bonds (1)	2			2		1